

SpectraCert: A Hierarchical System for Dynamic Spectrum License Enforcement

RYAN W. WEST, University of Utah, United States
DUSTIN MAAS, University of Utah, United States
DAVID M. JOHNSON, University of Utah, United States
JACOBUS VAN DER MERWE, University of Utah, United States

While various spectrum sharing approaches aim to remedy the spectrum crunch and improve spectral utilization, they typically support only radios that are certified for and physically locked down to a single frequency band and power configuration. This limits their utility and re-usability as they do not support flexible license changes post-production, but there is generally no alternative *dynamic* technical license enforcement mechanism suitable for more flexible SDRs (Software-Defined Radios). This limits the utility of SDRs: SDR operators must use unlicensed bands, or obtain special permission from the FCC via a manual process. Further, they are subject to fines if they operate outside these bounds and/or interfere with other spectrum users.

In response, we present SpectraCert—a dynamic, PKI-based (Public Key Infrastructure) system that enables safe, multi-faceted spectrum sharing while simultaneously enforcing license-holder rights directly within flexible transmitting radios. Entities receive specific frequency transmission rights from government authorities or other Spectrum-Authorized Entities (SAEs) through a *spectracert*: a certificate that specifies allowed frequencies and all required transmission conditions, such as allowed times, geographic areas, maximum transmit powers, expirations and more. Child spectracerts are derived from other spectracerts and have increasingly limited transmission rights, and *trusted modules* (TMs) within radios validate spectracert authenticity to ensure the rights defined by each parental spectracert are legal and not revoked. TMs only allow transmission according to validated spectracerts and control radio configurations. Spectracerts can also be used to verify SAE attestations of spectrum interference or occupancy reports. We implement a prototype of the SpectraCert system that automatically revokes or replaces spectracerts across SAE hierarchies when license updates occur, describe an integration with the OpenZMS spectrum management system, and evaluate how SpectraCert uniquely protects license rights while simultaneously allowing for more diverse, accessible spectrum sharing methods.

CCS Concepts: • **Networks** → **Wireless access networks**; **Network manageability**; **Network security**; • **Security and privacy** → **Digital rights management**.

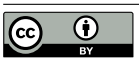
Additional Key Words and Phrases: spectrum sharing, license enforcement, spectral efficiency, PKI

ACM Reference Format:
Ryan W. West, Dustin Maas, David M. Johnson, and Jacobus Van der Merwe. 2025. SpectraCert: A Hierarchical System for Dynamic Spectrum License Enforcement. *Proc. ACM Netw.* 3, CoNEXT4, Article 33 (December 2025), 18 pages. https://doi.org/10.1145/3768980

1 Introduction

Radios communicate wirelessly using portions of the electromagnetic spectrum, which in general can only be occupied by a single transmitter at a time for a given geographical region [1]. Different parts of the spectrum (e.g., portions with non-overlapping center frequencies and bandwidths)

Authors' Contact Information: Ryan W. West, ryan@ryanwwest.com, University of Utah, United States; Dustin Maas, dmaas@cs.utah.edu, University of Utah, United States; David M. Johnson, johnsond@flux.utah.edu, University of Utah, United States; Jacobus Van der Merwe, kobus@cs.utah.edu, University of Utah, United States.



This work is licensed under a Creative Commons Attribution 4.0 International License.
© 2025 Copyright held by the owner/author(s).
ACM 2834-5509/2025/12-ART33
https://doi.org/10.1145/3768980

can be used concurrently by different transmitters within the same geographical region, but over time most *useful* spectrum for medium to long-range wireless communications (usually 0-6 GHz or Frequency Range 1 (FR1)) has become exclusively licensed or reserved for specific technologies. This "spectrum crunch" has made access to this range of spectrum increasingly limited and costly.

To combat this, various spectrum sharing systems have been designed and deployed which aim to improve *spectral utilization* (percentage of occupancy across all frequencies) by non-exclusively subleasing portions of spectrum for limited time intervals. Some examples include the FCC's Citizens Broadband Radio Service (CBRS), Canada's Non-competitive Local Licensing (NCL), Licensed Shared Access (LSA), and the Emerging Mid-Band Radar Spectrum Sharing (EMBRSS) investigation by the USA Department of Defense (DoD) [1–4]. While these efforts improve spectral utilization, they and spectrum licensing generally rely on locked-down certified radios to enforce spectrum owner license rights which must undergo lengthy, expensive, and manual processes to update or reconfigure. Governmental authorities such as the USA's FCC publish regulations that limit operation to be within the strict requirements of owners' spectrum licenses, which are primarily enforced via strict certification processes that cannot allow dynamic or temporary license changes.

In contrast, Software-Defined Radio (SDR) technology allows in theory for great configurability and flexibility in spectrum use. However, while it is widely used in radio systems, this flexibility is often curtailed because of inflexible licensing rules—many SDRs are nevertheless locked down in certified COTS RUs negating their benefits. For other SDRs, FCC and other entities regularly detect instances of owners operating at unauthorized frequencies or other disallowed configurations, which most spectrum sharing initiatives are similarly unequipped to handle [5, 6]. This malicious or accidental SDR spectrum misuse can cause degrading interference, disrupted connections, and overstepped license-holder rights, and is a major concern across both governments and academia. A notable exception is CBRS, whose Spectrum Access System (SAS) nodes issue temporary leases to base stations that are certified to halt transmission if an incumbent needs to begin transmitting in the area. While CBRS has been successfully deployed, its license enforcement mechanism is limited to its base stations within the 3.55-3.7 GHz band. Most other research focuses on detecting spectrum violations rather than preventing them in the first place.

In response, we present SpectraCert, a dynamic, hierarchical spectrum license enforcement system that enables safe and robust spectrum sharing. The SpectraCert system design adapts public key infrastructure (PKI) to create individual *spectracerts*, which are certificates that cryptographically grant a *spectrum-authorized entity (SAE)* transmission rights at a specific frequency range with specific conditions and restrictions. A spectracert may restrict the times, geographic areas, maximum power transmit levels, expiration dates, revocation check frequency, and many other radio operating parameters as required by the issuing SAE (or parent SAE). A spectracert is issued from one SAE with license rights to exactly one other SAE as demonstrated in Figure 1.

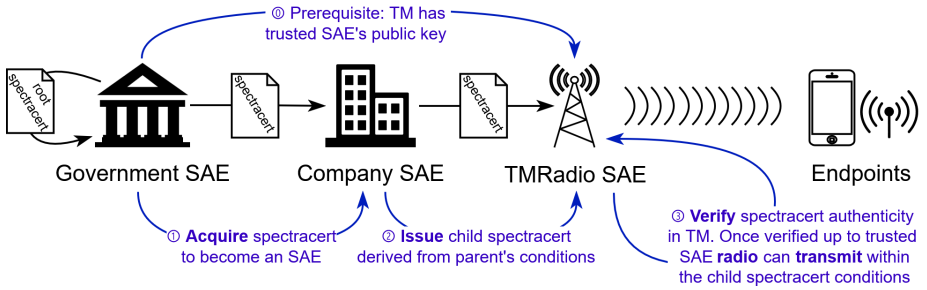


Fig. 1. SAEs and spectracerts

SAEs can include governmental entities such as the FCC or the United Kingdom's Ofcom, commercial mobile network operators (MNOs) such as AT&T, virtual MNOs, wireless research institutions, companies, neutral hosts, sub-entities within these entities, single radios, and more. Similar to the hierarchical nature of web-based PKI, a root of trust with ultimate spectrum authority (such as the NTIA or the FCC in the USA) creates a root spectracert with maximum transmission allowances and subsequently *derives* child spectracerts from this parent spectracert to all interested child entities, which can repeatedly further derive descendant spectracerts. A derived child spectracert is itself limited to the restrictions imposed on its parent spectracert but may also be more restricted.

Spectracerts *enforce* their encoded frequency rights and restrictions through the inclusion of a *trusted module (TM)* running in the software, firmware, and/or hardware of SDRs. A radio's TM prohibits any transmission that is not authorized by the spectracerts it has been issued, and the radio and TM together are called a TMRadio. The TM must verify a spectracert authenticity prior to usage, but unlike web-based PKI, it does not stop after simply verifying the signature(s) of all parental SAEs spectracerts. The TM additionally verifies that the frequencies and restrictions at *each level* in the hierarchical spectracert chain were properly authorized (e.g., if the FCC provided an MNO with a 20 MHz spectracert but the MNO then derived a child spectracert of 25 MHz to each of its radios, the TMRadios would reject them). Some spectracerts may also require the TM to make periodic revocation checks. Each SAE also runs a SpectraCert service with APIs to both request spectracerts and automatically revoke or replace them through all levels of the SAE hierarchy. For example, if FCC had issued a program experimental license (PEL) to a research institution but later needed to decrease the maximum transmit power of its radios, it could replace the original PEL-authorizing spectracert with a new one and all affected TMRadios would be automatically issued replacement derived spectracerts and could begin transmitting at a lower power.

Though this system, SpectraCert protects spectrum rights while simultaneously removing barriers to spectrum sharing—two benefits that are often mutually exclusive. Governments and manufacturers may be more willing to release new spectrum bands (EMBRSS) or more spectrum-flexible radios to the public with the assurance that radio operational parameters are checked [1]. By automating spectrum rights enforcement, SpectraCert could lower the barrier to entry of cellular and other radio fields without risking a 'wild west' of spectrum interference. Beyond radio license enforcement, SpectraCert also provides the capability to sign SAE attestations for radio monitoring, interference or occupancy reports, radio installation certification, and other events that would benefit from identification and proof of authorization.

We develop a prototype of the SpectraCert system running on real SDRs within the University of Utah's POWDER (Platform for Open Wireless Data-driven Experimental Research) platform [7]. We run SpectraCert sidecar services for several emulated SAEs in a CBRS-like setup to illustrate how different spectracert changes automatically affect radio transmit behavior via TMs. We integrate our SpectraCert implementation with OpenZMS [8], which automatically manages spectrum for outdoor cellular and non-cellular wireless experiments via POWDER-RDZ [9], and evaluate it.

We make the following contributions:

(1) We design a structure for spectracerts that specifies transmission rules, issuers and recipients, PKI properties, and extensions and allows deriving spectracerts.
(2) We design a protocol requiring SAEs and TMRadios to verify spectracerts up to the root SAE before transmitting or certification/identification usage. Spectracerts can be revoked to become unusable, and updates/revocations automatically propagate to affected SAEs/radios.
(3) We implement a prototype of SpectraCert with several SAEs/SDRs that illustrates use cases with a CBRS-like approach, and also integrate it with OpenZMS as part of the POWDER-RDZ, and evaluate how it dynamically allows transmission only according to license-holder rights.

2 Motivation and Related Work

There are two broad categories of technical spectrum license enforcement: *ex-post* (detecting unauthorized transmissions) and *ex-ante* (preventing unauthorized transmissions) [10, 11]. Weiss et al. provide a comprehensive analysis of general enforcement mechanisms covering both categories and apply it to spectrum and considers economic and technical impacts [11]. They stress that enforcement is critical for rights to afford any real value to their holders.

Ex-post works. The majority of license enforcement research focuses on *ex-post* approaches typically by sensing spurious transmissions and punishing violators. For example, CBRS requires environmental sensing capability (ESC) to detect potential incumbent transmissions and issue grant relinquishment messages to halt violating transmissions [2]. In terms of punitive action, Muthukumar et al. found that manual enforcement mechanisms of *ex-post* detection "operate[s] on a human time scale that is too slow to really be effective" [12] and discuss using adaptive 'jail sentences' where a malicious radio is forcibly disabled remotely if found to be operating in violation.

The need for ex-ante. However, SpectraCert follows an *ex-ante* methodology. While *ex-post* methods are necessary as no prevention method will be 100% effective, they typically require unscalable, manual effort. The FCC must often take action on SDR violators that cause spectrum interference due to the lack of proper prevention mechanisms[1] [5]. Bustamente et al. analyzed some of these spectrum violation records and predicts that violations will continue to increase with new SDRs and spectrum sharing methods, requiring automation to keep up [6].

It is therefore realistic and recommended to complement them with an *ex-ante* method to prevent most violations and avoid potentially damaging interference prior to detection and punishment [11, 13]. Indeed, the EMBRSS report concluded that "any final sharing framework will be flexible and employs different interference mitigation techniques" [1], In a similar vein, a 2019 NTIA Unlicensed National Information Infrastructure (U-NII) report shared various lessons learned related to mitigating harmful interference, such as "If installers and operators are physically able to illegally modify spectrum sharing devices to disable spectrum-sharing features, some will do so even knowing that they will be fined by regulators when they are eventually caught" [5]. They thus concluded that manufacturers should "restrict the ability to disable or modify their devices' spectrum sharing functionality as much as is physically possible", though radios are rarely restricted.

Ex-ante works. Wang et al. design and implement a system called FDMonitor that detects and immediately reports/halts unauthorized spectrum transmission transmission within POWDER wireless experiments [14]. It does this by pairing a dedicated Monitor SDR to listen to every Experimental SDR and ensure it is transmitting only within the spectrum grants assigned to it, and turns off the radio otherwise. This immediate detection makes this detection method fall somewhere in-between *ex-post* and *ex-ante*, though the system does not verify if the spectrum allocations it is enforcing came from a legitimate source. We later consider FDMonitor as one potential implementation of a Trusted Module (TM) to enforce spectracerts. Separately, Li et al. propose using virtualization and a secure radio middleware layer to allow secure SDR reconfigurations. This is reminiscent of an early SpectraCert TM, but focus on detection implementation rather than on validating the source of spectrum authority [15].

A related technology is the SIM card (Subscriber Identity Module) as used within user equipments (UEs). SIM cards are issued from the UE's carrier MNO and control the phone's cellular modem to operate within the bounds of that MNO. SIM cards contain profiles that specify the allowed operational parameters and are only modifiable by their issuing MNO [16]. SIM cards and modems are cryptographically secure and hardened to the point that even with rooted Android phones, it

[1]For example, for a growing list of FCC U-NII band violations, see https://www.fcc.gov/general/u-nii-and-tdwr-interference-enforcement

is extremely difficult to make unauthorized changes or even view settings, effectively preventing unauthorized transmissions. While SIM cards have been notably successful and are becoming increasingly dynamically deployable via embedded SIMs [17], they are only designed for *subscribers* in cellular networks, not general radio transmission. Furthermore, as only one SIM profile may be active at a time and the profile issuer solely determines its operational parameters, it is not possible to guarantee that the MNO will adhere to its spectrum licenses at a technical level. Neither is it likely the MNO would reconfigure customer SIM cards for those interested in additionally using other spectrum sharing schemas (e.g., CBRS base stations).

Currently commercially available SDRs have no corollary to dynamic SIM card transmission enforcement, which accentuates the current research focus on *ex-post* enforcement mechanisms. Many SDRs brands have not yet reached economy of scale keeping costs higher, and are not allowed by major cellular or other radio networks (typically, only locked-down certified radios are allowed), which limits utility [13]. The exception is CBRS, which requires usage of base stations known as CBRS Devices (CBSDs) that are certified to cease transmission within 60 seconds if required by their governing Spectrum Access System (SAS) [2]. This protects incumbent spectrum rights and has been seen as a success but is limited to the 3.55-3.7 GHz band. The spectrum sharing enforcement survey by Park et al. covers some earlier research that proposes using rule- or ontology-based policies radios prior to transmission [10]. Bahrak et al. further studied binary-based policies to decide when cognitive radios can transmit, but it is not generally applicable to all radios [18].

Works critical of today's ex-ante and ex-post methods. In the book *Spectrum Sharing: The Next Frontier in Wireless Networks*, chapter 17 covers policy enforcement in dynamic spectrum sharing that surveys some other related *ex-ante* approaches via device hardening, but specifically highlights the need for a language for spectrum access policies being a current challenge, as well as real-time enforcement of those policies [19]. It says: "Enforcing spectrum access control in legacy radios (e.g., cellular phones) is straightforward since the spectrum access policies are an inseparable part of the radio's firmware and platform. ... Unfortunately, the reconfigurability of SDRs/CRs not only makes them vulnerable to unauthorized modifications, but also makes it difficult to enforce spectrum policies.". Similarly, Woyach argues how spectrum regulations are difficult to change and that SDRs could be banned if not properly regulated, but clarifies that too much regulation may further decrease spectral utilization. Woyach suggests that a dynamic, remotely-configurable set of rules running on certified radios may be necessary to allow flexible regulation [13].

Many of these related works simultaneously point out open issues with technical spectrum license enforcement in tandem with their research. There is additionally no standard identity mechanism for license holders nor radios, as efforts have focused more on identifying radios through, e.g., power-based fingerprinting of modems [20] rather than providing technical identification upfront.

The DoD's EMBRSS 2023 Feasibility Assessment Report was released in response to an analysis to see if spectrum sharing could be introduced in the 3.1-3.45 GHz bands that are currently occupied by Federal incumbents and satellite research [1]. The analysis found that dynamic, time-based spectrum sharing that "expands and improves" CBRS is needed that allows for governmental emergency preemption and retains DoD as the regulatory primary. It additionally states that "DoD is concerned about the high possibility that non-Federal users will not adhere to the established coordination conditions at all times". On a related note, Berry et al. critiqued the NTIA's National Spectrum Strategy (NSS) in 2024, pointing out that spectrum realloactions have taken years or even decades longer than planned due to regulatory inefficiencies and has in some scenarios still released hundreds of MHz of spectrum less than promised [21]. They recommend that "the NSS should include mechanisms for encouraging easier applications and speedier implementation".

In summary, there are three problems in existing literature:

(1) More research is *ex-post* than *ex-ante*, making today's license enforcement less scalable and unable to prevent initial damaging interference. As the number of SDRs and parties interested in transmission grows, *ex-post* enforcement will grow more impractical.
(2) CBRS is the closest example of dynamic radio-based technical license enforcement to date, but is not used outside 3.55-3.7 GHz.
(3) Governments and companies want to improve spectral efficiency and protect spectrum rights, but have concerns with current SDRs and spectrum sharing initiatives that lack compliance guarantees, as these requirements are often opposing. Entities cannot automatically or dynamically deploy spectrum licenses that specify precise transmission requirements. Hesitation in initiatives such as EMBRSS indicates that these needs are not being met.

3 Design

3.1 SpectraCert's Main Components: SAEs and Spectracerts

The SpectraCert system at its core provides a protocol for certified communication using wireless spectrum under specific conditions. This has many applications, and one primary application or capability is to provide dynamic, preemptive licensing control to entities wishing to manage or transmit in wireless spectrum. Licenses traditionally originate from the government of the local jurisdiction with ultimate authority, such as NTIA (National Telecommunications and Information Administration) in the USA or OfCom in the United Kingdom. Some countries have additional governmental entities that manage specific use cases or ranges of spectrum, such as the FCC for commercial/academic usage and NTIA/DoD for federal/military usage. All of these are spectrum-authorized entities (SAEs), as are lower-level or *child* entities that receive more restricted radio authorizations, such as that of a single commercial MNO to operate in certain counties or a university to operate at certain times of day under low power levels. All types of SAEs also have a public and private keypair to allow for signing and verifying of spectracerts and related attestations.

Various additional levels of SAE are possible, such as temporary spectrum subleasing between MNOs or temporary grants to CBRS Priority Access License (PAL) holders, but an SAE will eventually transmit with an actual radio (also an SAE) with licenses. These licenses are transmitted between SAEs in the form of *spectracerts*, as illustrated in the example SAE hierarchy in Figure 2.

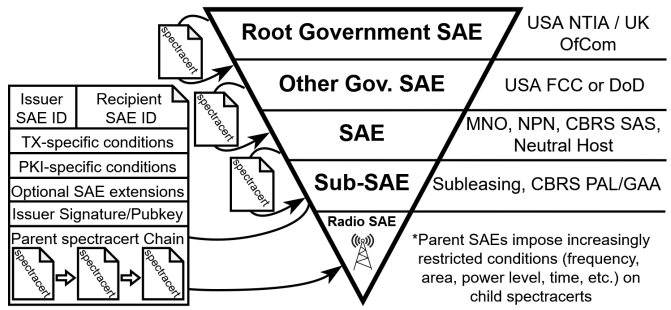


Fig. 2. Spectracert Structure within Example SAE Hierarchy

A spectracert is a cryptographic X.509 certificate sent from one parent/issuer SAE to one child/recipient SAE that specifies the allowed frequencies that the parent is authorizing the child to transmit at under specific conditions. A spectracert may contain various transmit-specific (TX) conditions that specify how transmissions may occur, including:

- **Frequency ranges**: A set of low-/high-frequency pairs at which the child SAE may transmit.
- **Geographic areas**: Locations permitting transmission.

- **Time ranges**: Periods in the day/week/month when transmission is permitted.
- **Maximum antenna transmit power, height/tilt**: The maximum equivalent isotropic radiated power (EIRP) or angle/height at which an antenna can transmit.
- **Allowed or disallowed hardware models/brands**: Certain SAEs (e.g., CBRS) may only allow a few approved radios to transmit while other SAEs (e.g., governmental) may choose to disallow known malicious or foreign brands or radio models.
- **Protocol restrictions**: SAEs may require certain frequency bands to only use time or frequency division duplexing (TDD/FDD), follow rules about listen-before-talk (LBT), require a specific wireless protocol, etc.
- **Protocol requirements**: Require certain behaviors, such as broadcasting identity codes in transmissions or tracking and reporting spectrum occupancy.

A spectracert may also specify various PKI-specific conditions, including:

- **Expiration timestamp**: Time after which the spectracert is no longer valid.
- **Maximum re-licensing depth**: Some SAEs may restrict their licensees from further sublicensing or delegating authority beyond their own radios.
- **Required online revocation check frequency**: How frequently a transmitting radio is required to check the revocation status of its spectracert.
- **Minimum trust level**: The minimum trust level of TM implementation on which this spectracert can operate.

Each spectracert is signed with the private key of its issuer SAE and also includes its public key, allowing the recipient SAE and anyone else to easily verify the creator's identity and detect modifications. As all spectracerts except the root are derived from other spectracerts, each spectracert also contains a list of IDs of every parent spectracert that was used in a derivation step up to the root spectracert. This list is critical: not only does it allow auditors to trace authority delegation to the root as in web-based PKI, but it *also* allows verification that conditions imposed by each upper SAE were correctly derived and upheld. This required signing and pairing of a spectracert to a specific TMRadio makes forging/replay attacks and other threats very difficult to carry out.

3.2 Technical License Enforcement Overview

While a spectracert by itself provides value by proving the spectrum authorizations of an SAE, *license enforcement at a technical level* requires an additional component, known as a *Trusted Module (TM)*. All radio SAEs contain a hardware- or software-based TM that permits and controls transmissions. A TM and radio together are called a TMRadio. A spectracert acts like a 'key' that the TM uses to unlock authorized transmission functionality if authenticated.

To verify a spectracert, the TM must first acquire all its parents up to a *trusted* spectracert, which is issued by a trusted SAE whose public key is initially embedded in the TM (normally the root spectracert) by the TM manufacturer (TMM) if one exists (similar to SIM cards). The TM first verifies that each condition in the base spectracert is either *as* restricted or *more* restricted than its immediate parent (e.g., a parent allowing transmission in the 2-2.1 GHz band that expires in one week could not create a valid derived spectracert allowing transmission in the 2-2.2 GHz band or that expires in one month). The base spectracert's recipient SAE ID must also match the parent spectracert's issuer SAE ID and have a valid signature. While some conditions are detectable and enforceable, such as lower and upper frequencies and expiration timestamps, other conditions such as antenna height and angle of tilt are difficult to enforce with firmware. CBRS has solved a similar problem by requiring certified installers to adhere to this physical data, and a spectracert could similarly inform installers of the required radio installation criteria which they certify.

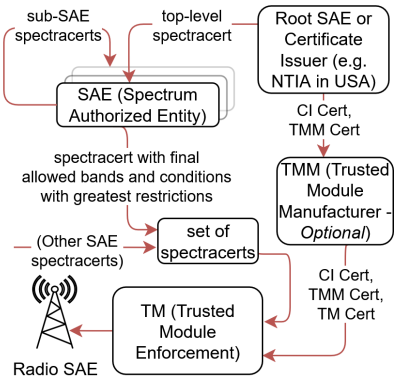


Fig. 3. SpectraCert Architecture

If all verification checks pass, the TMRadio sets the parent spectracert as the new base spectracert and sets *its* parent as the new parent spectracert, then repeats the process. Once a trusted parent spectracert has been reached and verified, the original base spectracert is considered verified.

Note that any SAE, not just a radio SAE, can also verify spectracerts in a similar manner, but only the TM uses that verification to reconfigure the radio to enforce transmissions according to the conditions. The TM must also periodically check revocation status for spectracerts that require it, and any revoked or expired spectracerts are invalidated and discarded.

Thus, the Spectracert system consists of an architecture as shown in Figure 3 that provides these steps for licensed radio transmission:

(1) **Acquire**: An entity acquires a spectracert from an upper SAE and becomes an SAE.
(2) **Issue**: The SAE decides which SAEs (including radios) can manage or transmit spectrum within its jurisdiction rights (i.e., conditions of its received spectracert), and how. When authorizing a radio/SAE, it issues them a new child spectracert derived from the parent.
(3) **Verify**: A TMRadio dynamically verifies all spectracerts issued to it up the hierarchy until it reaches a trusted parental spectracert. If verification passes, the TM reconfigures the radio to allow transmission within the validated spectracert bounds.

3.3 Trusted Module Details

TMRadios are not required for every spectracert capability—another notable capability is *attestation*. When an SAE creates a spectrum interference report or occupancy report intended for other SAEs, it can attach the accompanying spectracert that can be used to verify their authority. All SAEs have keys to sign spectracerts and reports. As a spectracert is simply composed of an X.509 certificate and JSON, it can be used flexibly for most existing PKI use cases and have their authorship and integrity independently verified without a TM. This may be used in proving compliance to military/governments or SAEs that have provided leased spectrum under agreed conditions.

Nevertheless, the SpectraCert system cannot actually *enforce* a spectracert's conditions at the radio parameter level unless a TM that can control the radio is present. Certifying a radio for legal transmission within a jurisdiction is currently expensive, difficult, and time-intensive, leading to limited certifications. But if a *TM* can be certified to only allow radio operation within validated spectracert parameters, many types of SDRs that integrate the TM may not need further certification.

It is likely that various TM implementations will exist, each with different *trust levels* with trade-offs between price and risk. A low trust level TM may be, e.g. software-only and may not require revocation checks, decreasing security but improving product compatibility and deployment/future

updates. MNO SAEs that sublease spectrum to one another may trust one another as it is in their best interest to avoid interference, so a cost-effective lower trust level may be appropriate. This low trust may also be appropriate for unlicensed spectrum running on low-power wireless hardware (e.g., small Wi-Fi routers) that cannot create significant harmful interference.

In contrast, a TM with a high trust level may require dedicated hardware such as a hardware security module (HSM) to verify spectracerts and securely reconfigure the radio, providing tamper-resistant device security at a higher cost. Security-conscious SAEs such as the Department of Defense may require that some of their issued spectracerts be operable only by some minimum trust level TM with hardened security hardware that are certified to, e.g., constantly check certificate revocation statuses and make spoofing time or other details very difficult. Checking certificate revocations necessitates an internet connection, and additional checks for clock synchronization could also be required to ensure an expired spectracert cannot be 'extended' beyond the true expiration time. As such, there may also be 'medium trust' levels that provide some out-of-band operation but still require an HSM, and some SAEs may choose to issue a complementary, limited but lower-trust spectracert that offers degraded use when a TM is offline. SAEs that require high trust level only spectracerts may find less prospective child SAEs less interested in the additional expense and inconvenience, but ultimately, the SAE is free to exercise their spectracerts' rights as they see fit. As with all other conditions, a spectracert that imposes a minimum trust level requires the same level (or higher) on all child SAEs that derive child spectracerts.

Trusted spectracerts are not necessarily limited to the root spectracert. In some instances, an intermediate SAE may wish to hide private spectracerts that it received from its parent SAEs. Without the complete chain of spectracert delegation, any TMs that choose to trust spectracerts issued by this non-root SAE must also assume that their authority is absolute and therefore need only verify spectracerts up to the trusted one. It may be difficult to find TMMs or certifiers willing to trust an intermediate SAE, especially at higher TM trust levels, but this workflow is possible.

While the spectracert structure is intended to cover many radio protocols and spectrum sharing architectures, it is impossible to anticipate every future regulation or rule. Thus, spectracerts may optionally contain an additional field for SAE extensions that can specify non-standard conditions—e.g., if a CBRS SAS anticipated future temporary required outages of a few minutes for incumbents that do not necessarily need to revoke a spectracert entirely, they could specify this behavior in an SAE extension. As many spectracert conditions *must* be enforced, a TM can only validate and use spectracerts if it has an accompanying extension module installed that can parse and enforce said SAE extension (provided by the same SAE that originally provided the extension).

TMRadio implementations may come in a variety of forms. Experimental approaches could follow the likes of FDMonitor, where transmitting radios are paired with a TM-controlled monitor radio that senses any attempt to begin operating out of spectracert range [14]. A higher trust level commercial TM may appear similar to the firmware that controls UE LTE/5G modems and allows transmission according to secure SIM card operational parameters. For lower trust, our POWDER prototype (explored more later) includes a TM wrapper around the OAI (OpenAirInterface) [22] and srsRAN [23] 5G gNB software stacks, which controls transmission by automatically modifying radio configuration files and restarting/halting the radio as spectracerts are updated/revoked. Certification of a particular TM implementation up to one of a set of standardized trust levels would require development a suite of rigorous tests similar to current FCC certification processes.

3.4 Automated Inter-SAE APIs

The SpectraCert System is named as such because it bundles the core spectracert and SAE components with HTTP Restful APIs that facilitate the management (issuance, revocation, replacement) of spectracerts between SAEs. Each SAE may run a small SpectraCert service designed to store

spectracerts and speak to other SAE Spectracert services via these APIs, or design their own. The APIs allow an SAE to retrieve spectracerts held by another SAE (including parent and child spectracerts) and request that the SAE consider creating a new child spectracert with specific conditions (and are described later on in Figure 6). It is of course up to every SAE to decide whether, how, and to whom to issue child spectracerts, and some SAEs may issue new spectracerts by default (e.g., an FCC SAE liberally grants spectracerts for unlicensed spectrum bands) while others may require manual processes or custom logic before issuing a new spectracert.

License-holder rights frequently change, and thus additional APIs are provided that allow revocation and replacement of spectracerts. TMs that require revocation checks do so by periodically checking a status URL run by parent SAE SpectraCert services and ensuring a non-revoked status is received. Additionally, when a parent SAE revokes a child spectracert, it may immediately inform affected child SAEs using another API that optionally allows sending a *replacement* spectracert with different conditions. An SAE could alternatively send a replacement child spectracert prior to revoking the original spectracert, depending on if downtime must be minimized as little as possible. As TMs check the authenticity and conditions derived from parental spectracerts up to a *trusted* spectracert, any revoked parent spectracert causes all of its child and descendant spectracerts to also be invalidated and revoked. The distributed SpectraCert system automatically propagates all API revocation requests from the original revoking parent SAE down to all descendant intermediate and child SAEs, including TMRadios that automatically stop exercising the rights of any newly revoked child spectracerts. The possibility of automatic mass revocations thus requires lightweight APIs with minimal response times. To avoid the case of a parent SAE not responding to spectracert revocation checks, a certified, TM-like module could be run at each SAE that could additionally allow the enforcement of maximum response times and API processing times at all SAEs.

If an SAE wishes to modify a spectracert (e.g. the FCC increases the maximum transmit power or reduces the frequency band of a Program Experimental License (PEL)), it does so by revoking the original spectracert and immediately issuing a nearly identical replacement spectracert with only specific conditions changed. The SpectraCert system can also auto-create and auto-issue replacement descendant child spectracerts all the way down to radio TMs to for widespread, automatic transmission reconfiguration. When creating a replacement child spectracert, the default derivation strategy is to leave all conditions the same unless the replacement parent is *more* restricted than the original revoked parent[2]. For example, if the revoked parent allowed transmission in the 1-4 GHz band, the revoked child 1-3 GHz, and the replacement parent 1-3 GHz, the replacement child is effectively unchanged. But if the replacement parent instead now allows 2-4 GHz, the replacement child must be constrained to 2-3 GHz. If the replacement parent was restricted to 3-4 GHz, a replacement child could not be automatically created. This strategy is by default applied to all conditions, but SAEs are free to use their own automatic or manual strategies instead.

4 Use Cases

The SpectraCert system is intended to broadly provide capabilities for as many stakeholders as possible, as illustrated in Figure 4. Use cases 1 and 2 show the cellular world, wherein MNOs may receive spectracerts from the FCC or other governmental SAE that grant spectrum rights in accordance with commercial spectrum auctions. Co-located MNOs may inter-lease subsets of their spectrum temporarily with each other to improve contiguity or correct unbalanced supply and demand [24]. Use case 3 shows unlicensed spectrum bands which may still require specific operating conditions such as Wi-Fi's maximum transmit powers at certain frequencies/routers [25], resulting in easily, automatically obtainable spectracerts that ensure said conditions are enforced.

[2]Intermediate SAEs could manually review newly granted conditions before including them in replacement child spectracerts.

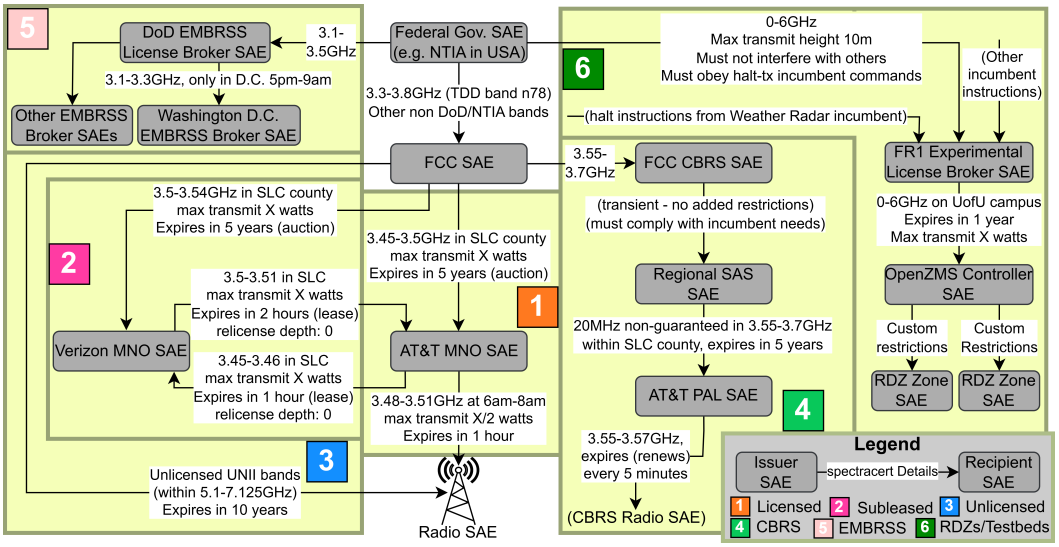


Fig. 4. Six Example Use Cases

Use case 4 exemplifies how CBRS SASes and PAL SAEs might interact if spectracerts were used to enforce their temporary 10-40 MHz allocations, which might be similarly applied to future EMBRSS deployments in use case 5. Finally, use case 6 proposes how research-driven PELs granted to universities could be distributed for individual experiments using OpenZMS, which we later demonstrate via prototype. Adoption of SpectraCert into these use cases would involve a gradual migration that may only affect the sale and deployment of future radios per use case.

5 Implementation

The SpectraCert system has been implemented on the University of Utah's POWDER Platform and tested over-the-air with X310/B210 SDRs [7] running OpenAirInterface [22] and srsRAN [23] 5G networks. Multiple SAE example implementations have been developed and tested in various outdoor multi-radio deployments. The Python-based SpectraCert service uses RESTful HTTP APIs to request, create, issue, revoke, and replace spectracerts and runs on a node for every SAE. We also implement an example low-trust level TM that wraps the SDRs and enforces validated spectracerts by reconfiguring the radio via configuration files and restarting or stopping the radio as necessary.

As SpectraCert is intended to provide flexible APIs that can integrate with any spectrum sharing service or protocol, we also describe and evaluate an integration of SpectraCert with OpenZMS [8]. The POWDER-RDZ platform infrastructure relies on OpenZMS[3] (Zone Management System), an automated spectrum management and intelligence system, that dynamically allocates spectrum grants to cellular and non-cellular wireless research experiments.

The combination of POWDER and OpenZMS is known as POWDER-RDZ (Radio Dynamic Zone) [9], and we have implemented and mapped several logical SpectraCert SAEs into POWDER-RDZ components as shown in Figure 5.

The Zone Management Controller (zms-zmc) service in OpenZMS is responsible for creating and managing spectrum *grants* based on delegated *spectrum* objects given to the OpenZMS deployment (such as FCC PEL frequencies allotted for POWDER-RDZ use), which operates with other ZMS services over gRPC as shown in the left of Figure 6. We have designed and implemented a new

[3]https://openzms.net/

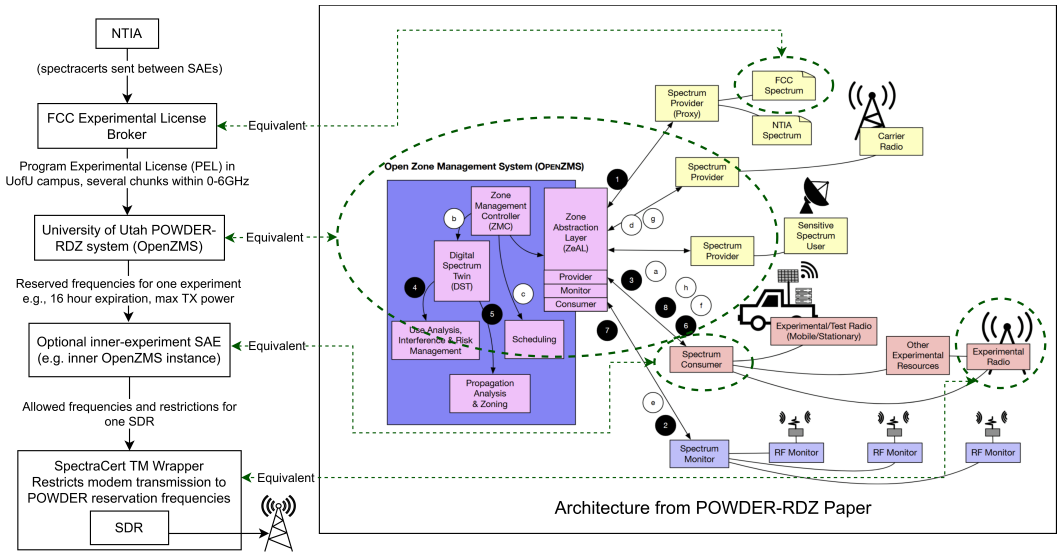


Fig. 5. Mapping of SpectraCert into POWDER-RDZ (dashed lines/ovals annotate equivalents)

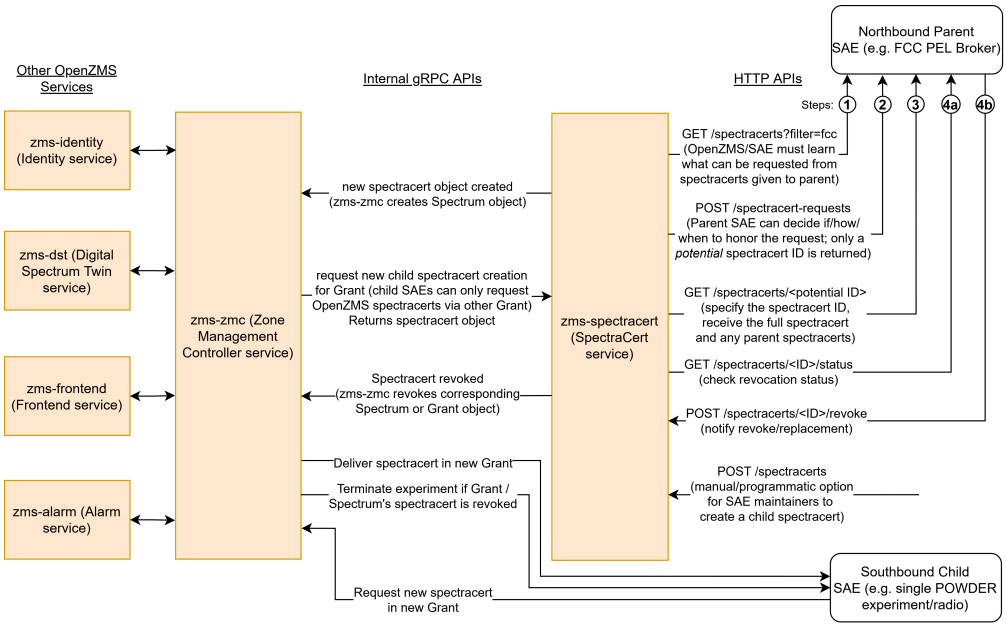


Fig. 6. SpectraCert HTTP APIs and their OpenZMS Integration

OpenZMS service, zms-spectracert, which handles SpectraCert HTTP API communications with parent SAEs externally, and notifies the zms-zmc service of newly created spectracerts or spectracert revocations. For each newly created spectracert, the zms-zmc service creates a new spectrum object matching the operational parameters of the spectracert, as shown in Figure 6. OpenZMS then uses existing scheduling and coexistence mechanisms to further delegate this spectrum object to consumers that request grants. When a new OpenZMS grant is created in a spectrum object authorized by a spectracert, OpenZMS requests a derived child spectracert from the zms-spectracert service, which is then associated with the OpenZMS grant—which is available to the end spectrum

user (and its TM) for verification. When OpenZMS receives a spectracert revocation notification, it revokes either the spectrum or grant object associated with the revoked spectracert. OpenZMS also accepts *claims* which preempt an existing grant for higher-priority transmission needs.

The spectracert HTTP APIs are spectrum sharing protocol- and system-agnostic, and are intended to work between any external SAEs to allow spectracert requests, creation, revocation, replacement, and other management. The common procedure to request and later potentially revoke a spectracert via these APIs is suggested in the top-right steps of Figure 6. In the POWDER-RDZ use case, OpenZMS creates a grant for each POWDER experiment, which contains an embedded per-experiment spectracert that can be used by experiment TMs to ultimately authorize transmission according to the grant and spectrum objects (which are themselves based on external parental spectracerts from for example, the FCC). Other spectrum sharing methods and SAEs can similarly use SpectraCert and interoperate with other SAEs however they wish (also including attestations of spectrum reports or validating reports/data from other alleged SAEs), as long as they integrate the HTTP APIs and TMs within radios verify spectracerts.

6 Evaluation

We illustrate the capabilities of the SpectraCert system via two scenarios that involve the creation, revocation, replacement, and transmission enforcement of spectracerts via SAE APIs on our implementation on POWDER radios and servers. The first shows inter-SAE CBRS-like spectracert operations for create/verify/replace/revoke procedures, and the second demonstrates the integration of SpectraCert with an existing OpenZMS SAE outdoor multi-radio operation.

6.1 CBRS-like Indoor OTA Evaluation

Figure 7 shows an over-the-air (OTA) scenario emulating a CBRS-like spectrum delegation hierarchy via several representative pseudo-SAEs each running a SpectraCert service[4]. In this example, the NTIA self-signs a root spectracert, giving it total spectrum authorities, and delegates the FR1 (0-6 GHz) band to the FCC via a child spectracert. FCC further designates the 3.55-3.7 GHz band for CBRS usage via a dedicated FCC-CBRS-like broker SAE (which may be separated similar to domains and subdomains). The broker subsequently grants an approved Spectrum Manager SAE the authority via another spectracert (similar to an FCC-approved SAS administrator [26]). As CBRS PAL auctions can be up to 40 MHz in bandwidth, the Spectrum Manager SAE initially issues a 40 MHz grant to a PAL-like Priority Access SAE via a spectracert, which exercises its spectrum rights by issuing a final spectracert to its local software-defined TMRadio. The TMRadio validates the chain of spectracert authority, then automatically begins transmitting at the full 40 MHz band as shown in the spectrograms on the right of Figure 7. Data was captured via the UHD utility (USRP Hardware Driver) from a monitor SDR while sending a high rate of iPerf3 packets from a UE to the TMRadio running an OAI gNB to fill all data Physical Resource Blocks (timing is not to scale).

In the middle section of Figure 7, we next illustrate the SpectraCert revocation with replacement APIs by considering recent USA reports that the entire CBRS band could be reassigned as more spectrum is released.[5] Consider a hypothetical future in which the CBRS band is migrated from 3.55-3.7 GHz to 3.4-3.55 GHz, and all existing CBRS base stations need to switch frequencies (another frequency range could also easily be used). SpectraCert accomplishes this when the FCC initially replaces the CBRS-like Broker's spectracert with a new one with downshifted frequencies. The broker then revokes the original spectracert, adds its replacement, and then must create a new child spectracert for the SAS-like Spectrum Manager SAE that replaces the original child spectracert.

[4]This system is only designed to be hierarchically *similar* to CBRS and does not use actual certified CBSDs or SAS software.
[5]See https://www.lightreading.com/5g/trump-i-am-going-to-free-up-plenty-of-spectrum-for-auction-

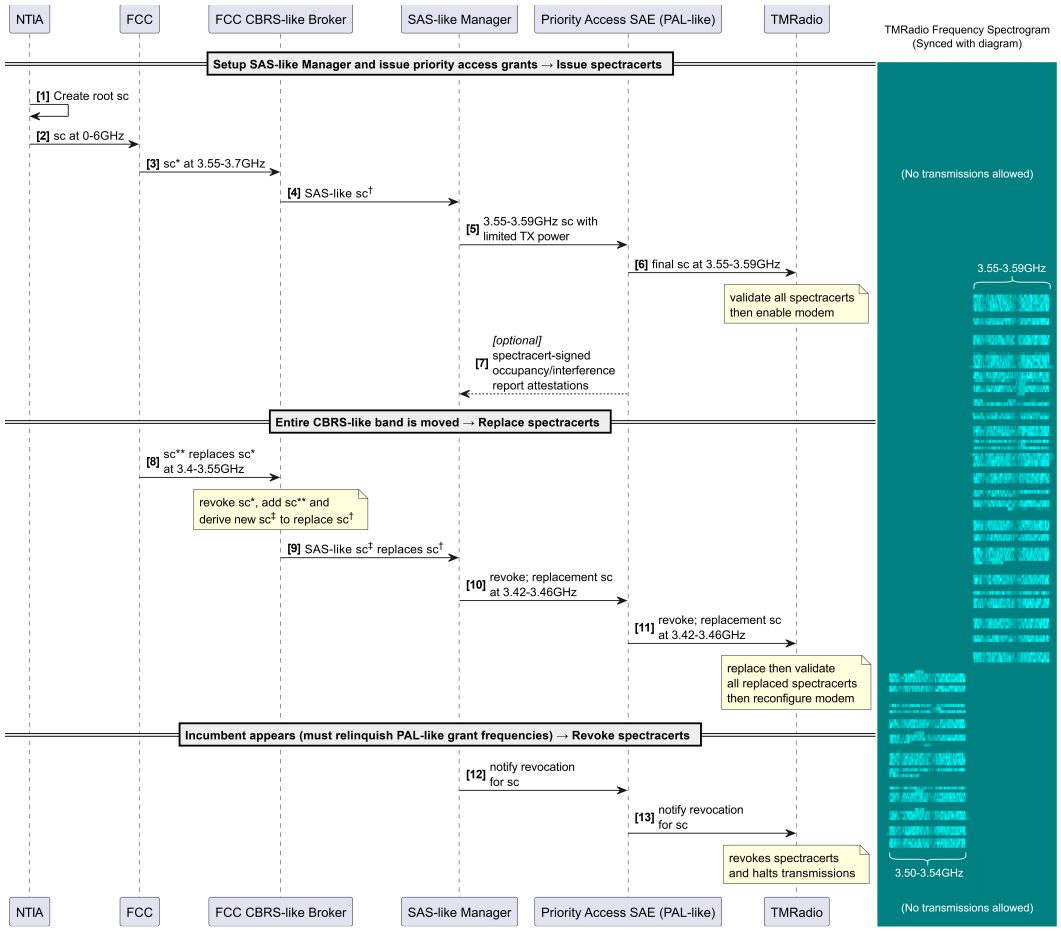


Fig. 7. CBRS-like usage of spectracerts (sc) to automatically adjust and enforce SDR transmission rights

The SAS-like Spectrum Manager then reassigns the Priority Access SAE a new 40 MHz band (e.g., 3.50-3.54 GHz) in a replacement spectracert which is used to create another child spectracert sent to the TMRadio. The TMRadio can no longer use the revoked spectracert and validates the replacement spectracert, then reconfigures the radio to transmit 40 MHz at the lower frequencies. We have also tested cases where the replacement spectracert is reduced to a 20 MHz bandwidth, which produces a 20 MHz spectrograph as expected. This process is easily automated if each SAE allows creation of child spectracerts that simply downshift the frequencies of the original child, which occurs here.

Finally, the lowest section of Figure 7 demonstrates how spectrum rights are revoked when a CBRS-like incumbent appears. When a SAS detects an incumbent presence (via, e.g., ESC), it may need to relinquish the grants of GAA users and later PAL users to prevent primary interference. A similar process is demonstrated here where Spectrum Manager SAE revokes the replacement spectracert from the Priority Access SAE which subsequently revokes the now-invalid replaced child spectracert. As the TMRadio no longer has any valid spectracerts dedicated to it, all transmissions are halted. The same behavior occurs when the TMRadio itself detects that a spectracert expiration timestamp has passed, which it checks every other second by default, or if the TMRadio uses a spectracert that requires active revocation checks and it learns of a new revocation status.

New advancements in CBRS policy allow for GAA users that lack interference protection to use GAA Coordination Areas (GCAs) to improve co-existence and reduce interference. GAAs can share information with each other and their SAS to allow careful reselection of GAA channels and potentially synchronizing other operational timing parameters [27]. A GCA-enabled SAS can act on GAA interference and occupancy reports to automatically revoke and replace spectracerts via APIs whenever it reorganizes CBRS frequency allocations (as represented by step 7 in Figure 7). As each spectracert contains a valid X.509 certificate, PAL and GAA SAEs can trivially sign report data which the SAS can use to easily authenticate the report origin.

The spectracert processes illustrated in our CBRS-like system apply generally as well. While the CBRS scenario demonstrates the TMRadio constantly exercising its full spectrum rights, real deployments might only use portions of their allowances at certain times to match customer demand and use case requirements. If any SAE attempts to create a spectracert that exceeds the limits of its parent, such as specifying a higher transmit power/bandwidth or ignoring re-licensing depth, lower SAEs and/or the radio's TM immediately detects and rejects the spectracert. The same is true for modifying existing spectracerts—if the FCC CBRS-like broker SAE tries to increase CBRS to 3.5-3.8 GHz without the proper parent spectracert, cryptographic signature verification fails at the Spectrum Manager SAE and the TM (which treats the root NTIA spectracert as trusted) cannot validate any descendant spectracerts derived from the modified spectracert.

Timing metrics are found in Table 1, where this entire process completed in under a quarter second, and most individual spectracert operations finish in under 14ms, tested with hundreds of stored spectracerts and chains of a dozen descendant spectracerts. These dynamic APIs are several orders of magnitude faster than current FCC license certification processes (e.g., weeks or months).

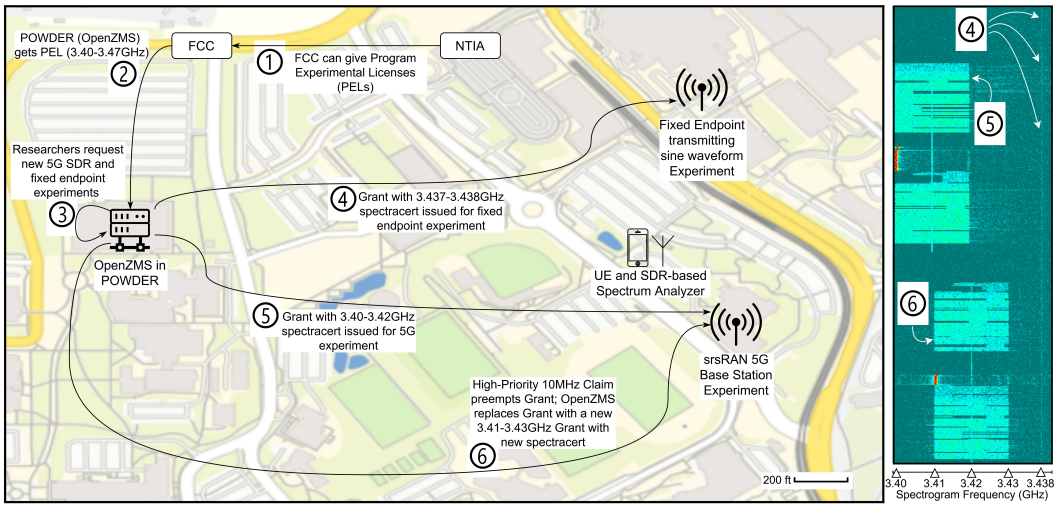


Fig. 8. OpenZMS Running Two Outdoor Radio Experiments Transmitting By Spectracert Validation

6.2 OpenZMS-Enabled Multi-Experiment Outdoor OTA Evaluation

We next demonstrate an end-to-end outdoor OTA scenario involving a SpectraCert-integrated OpenZMS instance and multiple cellular and non-cellular radios transmitting. Figure 8 shows part of the POWDER testbed on the University of Utah campus including a fixed endpoint with Intel NUC and Ettus B210 on the top-right; an Intel NUC and Ettus B210 with C-band TDD RF front end running an srsRAN 5G base station on the bottom right; a Motorola Edge 5G handset that attaches to the 5G base station between the fixed endpoint and base station; a laptop and Ettus X310 for

Table 1. SpectraCert Performance Metrics

Timing Metric Description	Time
Time for a child spectracert to be derived from a parent spectracert	2.9 ms
Round-trip time for API requests (requesting, getting, revoking spectracerts)	9 ms to 13 ms
Local runtime of full CBRS-like scenario (skipping radio transmission)	233 ms
OpenZMS step 6: Replacement Grant with spectracert arrive at radio node	5.7 s
Spectracert verification process duration in the TM	3.0 ms
OpenZMS step 6: Time to start radio transmission process post-verification	<1 ms
OpenZMS step 6: Claim submission to spectracert-verified transmission start	7.7 s

capturing a spectrogram of the RF activity of interest co-located with the UE; and compute nodes running the Open5GS core network and OpenZMS within the POWDER datacenter on the left. Additional compute nodes run other SAE SpectraCert services. The process includes six key steps:

(1) An emulated NTIA SAE gives an emulated FCC SAE authority to issue PELs via a spectracert.
(2) FCC issues an example 70 MHz PEL via a spectracert to POWDER for wireless experiments.
(3) POWDER, through OpenZMS, creates a corresponding 70 MHz Spectrum object that it uses to allocate portions of bandwidth to different researchers that run short-term experiments.
(4) When an experiment using the upper fixed endpoint is started, OpenZMS creates a new Grant at 3.438 GHz with an embedded spectracert, which is verified by the experiment TMRadio to allow transmission of a sinusoidal continuous waveform. This is shown in the faint, thin vertical line of activity going down the far right of Figure 8's spectrogram.
(5) A separate but concurrent experiment begins which receives another 20 MHz grant with a 3.40-3.42 GHz embedded spectracert that is used to run a 5G network with an srsRAN base station (the small streaks of red activity at 3.40 GHz show a co-located, connected UE transmitting in uplink back to the base station).
(6) An OpenZMS mechanism is invoked that allows users to submit a high-priority claim which takes precedence over an existing experiment grant. When a claim is submitted, OpenZMS lets the experiment receive a replacement grant that relocates spectrum to a new available center frequency if one exists. We submit a 10 MHz claim at 3.4-3.41 GHz which causes the old grant to be revoked and replaced with a new 3.41-3.43 GHz grant and new spectracert that is then verified. This allows transmission to resume shifted up 10 MHz. The fixed endpoint experiment's spectracert and transmission are unaffected. This entire process takes less than 10 seconds as shown in Table 1 and is mostly due to existing OpenZMS operations, with spectracert verification taking only 3 ms. This timing falls well under CBRS's 60-second time limit in which a CBSD must shut off due to an incumbent grant relinquishment notification.

7 Conclusion

This paper presents SpectraCert, a system that enables dynamic, technical enforcement of spectrum licenses via PKI-based spectracert certificates spanning from the government authority down to individual radios. We detail the structure used to specify spectracert frequency rights and restrictive conditions, and subsequently define the APIs and processes needed for SAEs to create, utilize, and revoke spectracerts to prove spectrum ownership and enforce authorized radio transmissions. Our POWDER implementation and evaluation illustrate how SpectraCert can be concurrently applied across a broad variety of new and existing use cases. We hope that SpectraCert inspires governments, standards committees, and companies to explore more dynamic spectrum licensing options and SDRs with the assurance that resulting delegated rights can be enforced.

Acknowledgments: This work was supported by the NSF through the following grants: #2232463, #2346555, and #2431961.

References

[1] "EMBRSS: Emerging Mid-Band Radar Spectrum Sharing Feasibility Assessment Report," 2023. [Online]. Available: https://dodcio.defense.gov/Portals/0/Documents/Library/DoD-EMBRSS-FeasabilityAssessmentRedacted.pdf

[2] "Requirements for Commercial Operation in the U.S. 3550-3700 MHz CBRS," 2024. [Online]. Available: https://winnf.memberclicks.net/assets/CBRS/WINNF-TS-0112.pdf

[3] Government of Canada, "Consultation on a non-competitive local licensing framework, including spectrum in the 3900-3980 mhz band and portions of the 26, 28 and 38 ghz bands," Aug. 2022, sPB-003-22. Accessed: 2025-06-04. [Online]. Available: https://ised-isde.canada.ca/site/spectrum-management-telecommunications/en/learn-more/key-documents/consultations/consultation-non-competitive-local-licensing-framework-including-spectrum-3900-3980-mhz-band-and

[4] M. Matinmikko, H. Okkonen, M. Palola, S. Yrjola, P. Ahokangas, and M. Mustonen, "Spectrum sharing using licensed shared access: the concept and its workflow for lte-advanced networks," *IEEE Wireless Communications*, vol. 21, no. 2, pp. 72–79, April 2014.

[5] F. H. Sanders, "Lessons Learned from the Development and Deployment of 5 GHz Unlicensed National Information Infrastructure (U NII) Dynamic Frequency Selection (DFS) Devices," 2019.

[6] P. Bustamante, D. Das, J. S. Rose, M. Gomez, M. B. H. Weiss, J.-M. Park, and T. Znati. (2020) Toward Automated Enforcement of Radio Interference. [Online]. Available: https://papers.ssrn.com/abstract=3749751

[7] J. Breen et al., "POWDER: Platform for Open Wireless Data-driven Experimental Research," in *14th International Workshop on Wireless Network Testbeds, Experimental Evaluation & Characterization (WiNTECH '20)*, 2020.

[8] D. Johnson, D. Maas, S. Tadik, A. Orange, L. Stoller, K. Webb, M. B. I. Awan, J. Bills, M. Gomez, A. Sarbhai, G. Durgin, S. Kasera, N. Patwari, D. Schurig, and J. V. der Merwe, "Building Radio Dynamic Zones with OpenZMS," *IEEE Transactions on Cognitive Communications and Networking*, pp. 1–1, 2025.

[9] D. Johnson, D. Maas, S. Tadik, A. Orange, L. Stoller, K. Webb, M. B. I. Awan, J. Bills, M. Gomez, A. Sarbhai, G. Durgin, S. Kasera, N. Patwari, D. Schurig, and J. Van der Merwe, "Powder-RDZ: Prototyping a Radio Dynamic Zone using the POWDER platform," in *2024 IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN)*, 2024, pp. 69–78. [Online]. Available: https://ieeexplore.ieee.org/document/10632848

[10] J.-M. Park, J. H. Reed, A. A. Beex, T. C. Clancy, V. Kumar, and B. Bahrak, "Security and Enforcement in Spectrum Sharing," vol. 102, no. 3, pp. 270–281, 2014. [Online]. Available: https://ieeexplore.ieee.org/document/6732887/?arnumber=6732887

[11] M. B. Weiss, W. Lehr, M. Altamaimi, and L. Cui, "Enforcement in Dynamic Spectrum Access Systems," *SSRN Electronic Journal*, 2012, http://www.ssrn.com/abstract=2032242.

[12] V. Muthukumar and A. Sahai, in *Fundamental Limits on Ex-Post Enforcement and Implications for Spectrum Rights | IEEE Conference Publication | IEEE Xplore*. IEEE. [Online]. Available: https://ieeexplore-ieee-org.ezproxy.lib.utah.edu/abstract/document/7920755

[13] K. A. Woyach, A. Sahai, G. Atia, and V. Saligrama, "Crime and Punishment for Cognitive Radios," in *2008 46th Annual Allerton Conference on Communication, Control, and Computing*. IEEE, pp. 236–243. [Online]. Available: http://ieeexplore.ieee.org/document/4797562/

[14] J. Wang, J. Gornet, A. Orange, L. Stoller, G. Wong, J. V. d. Merwe, S. K. Kasera, and N. Patwari, "Two Measure is Two Know: Calibration-free Full Duplex Monitoring for Software Radio Platforms," in *2024 IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN)*, pp. 267–276. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10632795

[15] C. Li, A. Raghunathan, and N. K. Jha, "An architecture for secure software defined radio," in *Automation & Test in Europe Conference & Exhibition 2009 Design*, 2009, pp. 448–453. [Online]. Available: https://ieeexplore.ieee.org/document/5090707

[16] *Embedded SIM Remote Provisioning Architecture Version 4.3*, Std., 2022. [Online]. Available: https://www.gsma.com/solutions-and-impact/technologies/esim/wp-content/uploads/2022/11/SGP.01-v4.3.pdf

[17] D. G. Koshy and S. N. Rao, "Evolution of SIM Cards – What's Next?" in *2018 International Conference on Advances in Computing, Communications and Informatics (ICACCI)*, pp. 1963–1967. [Online]. Available: https://ieeexplore.ieee.org/document/8554774/citations?tabFilter=papers#citations

[18] B. Bahrak, A. Deshpande, M. Whitaker, and J.-M. Park, "BRESAP: A Policy Reasoner for Processing Spectrum Access Policies Represented by Binary Decision Diagrams," in *2010 IEEE Symposium on New Frontiers in Dynamic Spectrum (DySPAN)*, pp. 1–12. [Online]. Available: https://ieeexplore.ieee.org/document/5457867

[19] J. (Jerry) Park, V. Kumar, and T. Oyedare, "Policy Enforcement in Dynamic Spectrum Sharing," in *Spectrum Sharing: The Next Frontier in Wireless Networks*. IEEE, 2019, pp. 341–359. [Online]. Available: https://ieeexplore.ieee.org/document/9080515

[20] C. R. Aguayo González and J. H. Reed, "Power fingerprinting in SDR integrity assessment for security and regulatory compliance," vol. 69, no. 2–3, pp. 307–327, 2011. [Online]. Available: https://doi.org/10.1007/s10470-011-9777-4

[21] M. Honig, R. Berry, and T. W. Hazlett, "Evaluating the National Spectrum Strategy." [Online]. Available: https://www.ssrn.com/abstract=4913586
[22] F. Kaltenberger, G. d. Souza, R. Knopp, and H. Wang, "The openairinterface 5g new radio implementation: Current status and roadmap," in *WSA 2019; 23rd International ITG Workshop on Smart Antennas*, 2019.
[23] Software Radio Systems, "srsran/srsRAN_Project: Open source O-RAN 5G CU/DU solution from Software Radio Systems (SRS)," https://github.com/srsran/srsRAN_Project, Jun. 2025.
[24] R. W. West, M. B. I. Awan, M. Gomez, V. Gopalakrishnan, D. Maas, A. Sivakumar, J. Van der Merwe, and I. Zelaya, "Spectraleas: Dynamic Wireless Spectrum Subleasing," 2025, pp. 355–364, in *20th IEEE International Symposium on Dynamic Spectrum Access Networks*.
[25] "Operation within the bands 902-928 MHz, 2400-2483.5 MHz, and 5725-5850 MHz," Code of Federal Regulations, U.S. Government Publishing Office (GPO), 2023, 47 C.F.R. §15.247. Current as of May 5, 2025.
[26] FCC, "Less telecommunications bureau and office of engineering and technology approve four spectrum access system administrators for full scale commercial deployment in the 3.5 ghz band and emphasize licensee compliance obligations in the 3650-3700 mhz band under part 96," 2020. [Online]. Available: https://docs.fcc.gov/public/attachments/DA-20-110A1.pdf
[27] O. Alliance, "Best Practices for Collaborative GAA Coexistence," OnGo Alliance, 2024. [Online]. Available: https://ongoalliance.org/wp-content/uploads/2024/10/GAA-Coexistence-Whitepaper-1.0.pdf

Received June 2025; accepted September 2025